

Wonkyo Choe

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Professional Summary

Ph.D. Candidate in Computer Science specializing in system infrastructure for large-scale machine learning. My dissertation builds efficient, scalable infrastructure for serving agentic and generative AI workloads, designing systems that coordinate compute, memory, and storage across heterogeneous resources. I work across the full stack, from operating systems and virtualization to LLM and diffusion inference, with published systems research spanning memory management, edge inference, and model compression.

Education

University of Virginia, Ph.D. in Computer Science Advisor: Felix Xiaozhu Lin	2021 – 2027 (Expected)
Ajou University, M.Eng in Computer Engineering (With Full Scholarship)	2018 – 2021
Ajou University, B.Eng in Computer Engineering	2011 – 2018

Industry Experiences

Web Development Intern Biztech Inc, Irvine, CA	Feb 2017 – Feb 2018
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- Engineered end-to-end web applications for clients; managed the full development lifecycle from initial front-end design to backend architecture and post-launch support; Funded by university international scholarship.

Selected Publications

SpMAP: Transparent Sparsity for LLMs, MobiSys'26 [\[PDF\]](#)
Wonkyo Choe, Felix Xiaozhu Lin. Syscall-driven sparse LLM loading; cut resident memory by 63% on edge devices.

RWKV-Lite: Deeply Compressed RWKV for Resource-Constrained Devices, arXiv 2025 [\[PDF\]](#)
Wonkyo Choe, Yangfeng Ji, Felix Xiaozhu Lin. SVD-based compression; up to 5× smaller memory footprint.

Proto: A Guided Journey through Modern OS Construction, SOSP'25 [\[PDF\]](#)
Wonkyo Choe*, Rongxiang Wang*, Afsara Benazir*, Felix Xiaozhu Lin. Educational OS with progressive apps (* equal contribution).

AnA: An Attentive Autonomous Driving System, ASPLOS'25 [\[PDF\]](#)
Wonkyo Choe, Rongxiang Wang, Felix Xiaozhu Lin. Dual-perception modules; 2× fewer collisions, 43% less compute.

Skills

Languages: C, C++, Python3, Bash, Latex, Markdown
Skills: LLM Training/Inference, Diffusion Inference, Linux Kernel Development, System optimization, Performance Profiling, GPU Computing (CUDA)
Technologies: PyTorch, Ray, SGLang, Docker, K8s, Git

Honors & Awards

Departmental Scholarship, Ajou University (2011, 2016, 2017, 2018)
Graduate Outstanding Teaching Award, University of Virginia (2025)

Professional Activities

Peer Reviewer, IEEE Transactions on Mobile Computing (2024)